eDNA – lab protocol
BIO779P

DNA Extraction (Soil)

Based on the Qiagen DNeasy Blood & Tissue Kit

Goal: Extract high-quality environmental DNA for downstream PCR and sequencing.

• Wear gloves at all times.

This protocol describes the extraction and purification of total genomic DNA from soil, using a modified protocol of the DNeasy Blood & Tissue Kit.

Preparations before starting

- Buffers **ATL** and **AL** may form precipitates during storage. If this occurs, warm to **56 °C** until fully dissolved.
- Buffers **AW1** and **AW2** are supplied as concentrates. Before first use, add the required volume of **96–100% ethanol** as indicated on the bottle.
- Preheat a thermomixer, shaking water bath, or rocking platform to **56 °C**.

1. Sample preparation

☐ Weigh **5g of soil** and add it to the tube.
☐ Add **250 µl Buffer ATL** to the tube.

2. Tissue lysis

☐ Add **20 µl Proteinase K.**
☐ Mix thoroughly by vortexing.
☐ Incubate at **56 °C – ~~30 min~~** (or longer) 2 hours.

Mix occasionally during incubation by vortexing or gentle agitation.

3. Binding preparation

☐ Vortex the lysate for **15 seconds.**
☐ Add **200 µl Buffer AL** and mix thoroughly by vortexing.
☐ Add **200 µl ethanol (96–100%)** and mix thoroughly again. → Centrifuge 16,000 RCF → 1 min

A white precipitate may form.

4. DNA binding

☐ Transfer the entire mixture (including any precipitate) to a **DNeasy Mini spin column** placed in a **2 ml collection tube.**
☐ Centrifuge for **1 minute at ≥6000 × g (≈8000 rpm).**
☐ Discard the flow-through and collection tube.

5. Wash 1

☐ Place the spin column into a new **2 ml collection tube.**
☐ Add **500 µl Buffer AW1.**
☐ Centrifuge for **1 minute at ≥6000 × g.**
☐ Discard the flow-through and collection tube.

6. Wash 2 and membrane drying

☐ Place the spin column into a new **2 ml collection tube.**
☐ Add **500 µl Buffer AW2.**
☐ Centrifuge for **3 minutes at 20,000 × g (≈14,000 rpm)** to dry the membrane.

Complete drying is essential, as residual ethanol can inhibit downstream applications. If ethanol carryover suspected, perform an additional 1-minute centrifugation.

7. DNA elution

☐ Transfer the spin column to a clean **1.5 ml or 2 ml microcentrifuge tube.**
☐ Add **200 µl Buffer AE** directly onto the membrane.
☐ Incubate at room temperature for **1 minute.**
☐ Centrifuge for **1 minute at ≥6000 × g** to elute DNA.

Eluting in **100 µl** instead of 200 µl will increase DNA concentration but reduce total yield. **Note:** Do not elu more than **200 µl** into a 1.5 ml tube, as the column may contact the eluate.

2. Visualising Native Genomic DNA on a 1% Agarose Gel

Purpose: To assess the integrity, size distribution and purity of extracted genomic DNA (e.g., soil DNA) under non-denaturing conditions.

Reagents and Materials

- Agarose powder
- 1× TAE buffer (or TBE, but TAE gives slightly better migration for long DNA)
- GelRed DNA stain
- DNA loading dye (glycerol-based)
- High-range DNA ladder (e.g., 0.5–20 kb)
- Nuclease-free water

- Gel tray, casting platform, combs
- Power supply & electrophoresis chamber
- Microwave or hot plate
- Pipettes & filtered tips
- Blue-light transilluminator or UV (blue light preferred for gDNA)
- 1.5 ml tubes

Prepare a 1% SYBR™ Safe Agarose Gel

☐ 1. Weigh agarose: 1.0 g agarose per 100 ml 1× TAE.
☐ 2. Dissolve agarose:
Add agarose to 100 ml TAE in an Erlenmeyer flask / microwave-safe bottle.
Heat in microwave in 20–30 s bursts, swirling between intervals, until solution is clear.
☐ 3. Cool to ~60 °C Comfortable hand-warm. Too hot and SYBR™ Safe will degrade.
☐ 4. Add SYBR™ Safe: Add 10 µl SYBR™ Safe per 100 ml gel (1:10,000 final). Swirl gently to mix, avoid bubbles.
☐ 5. Pour gel: Place the comb into the casting tray. Pour the agarose slowly into the tray. Allow to set for 20–30 min at room temperature until firm and opaque.

Prepare DNA Samples and Ladder

Native genomic DNA often lacks dye, so:

- Mix 5 µl DNA with 1 µl 6× loading dye.
- If concentration is low, load up to 10 µl (keeping dye at 1× final).

Recommended loading amount:

- 100–300 ng genomic DNA per lane
  (you'll see a high-molecular-weight smear near the top).

Ladder preparation

- Load 1–5 µl depending on concentration.

Loading the Gel

☐ Place the gel in the electrophoresis chamber and cover with 1× TAE.
☐ Keep wells at the negative (black) electrode side.
☐ Carefully pipette into each well:

- Ladder in the first well
- DNA samples in subsequent wells
- Lower the pipette tip just below the buffer surface to avoid tearing the well.

Electrophoresis

1. Attach the lid and connect electrodes.
2. Run gel at 100 V.

Run for 45–60 minutes, or until the dye front has migrated 2/3 down the gel.

Visualisation

1. Gently lift the gel and place onto a blue-light transilluminator.
2. View through a filter.
3. Capture an image using the gel documentation system or camera.

You results:

Interpretation

High-quality native DNA

- *Bright, dense band near the top of the gel (≥20 kb).*
- *Minimal smearing.*

Moderately sheared DNA

- *Smear extending downwards from a high-molecular-weight zone.*
- *Expected with soil extractions.*

Extensive degradation

- *Broad smear extending well into lower molecular-weight region.*

RNA contamination

- *Bright fuzzy bands at the bottom (100–500 bp).*

Salt/protein contamination

- *DNA may appear stuck in wells or show distorted migration.*